Appendix. User guide to the intelligent MWD-data analysis web platform

1.1 Platform overview

The intelligent MWD-data analysis platform is a web application that packages the method of the main text as an online service, at https://mwd.drcui.top. The browser is the only client; the back end builds a RESTful API and WebSocket services on Python FastAPI, and the front end uses Vue 3 with Element Plus. Users complete the whole workflow, from uploading raw data to obtaining lithology-identification output, without installing any client or writing code. The latest versions of Chrome, Firefox or Edge are recommended, at a screen resolution of at least 1280×720.



Fig. 1. Platform home page.

New users register themselves on the sign-in page; access is granted after an administrator approves the request. Signing in leads to the dashboard, which carries summary cards for datasets and tasks at the top, shortcuts covering the whole workflow in the middle, and a list of recent tasks at the bottom that refreshes every two seconds.

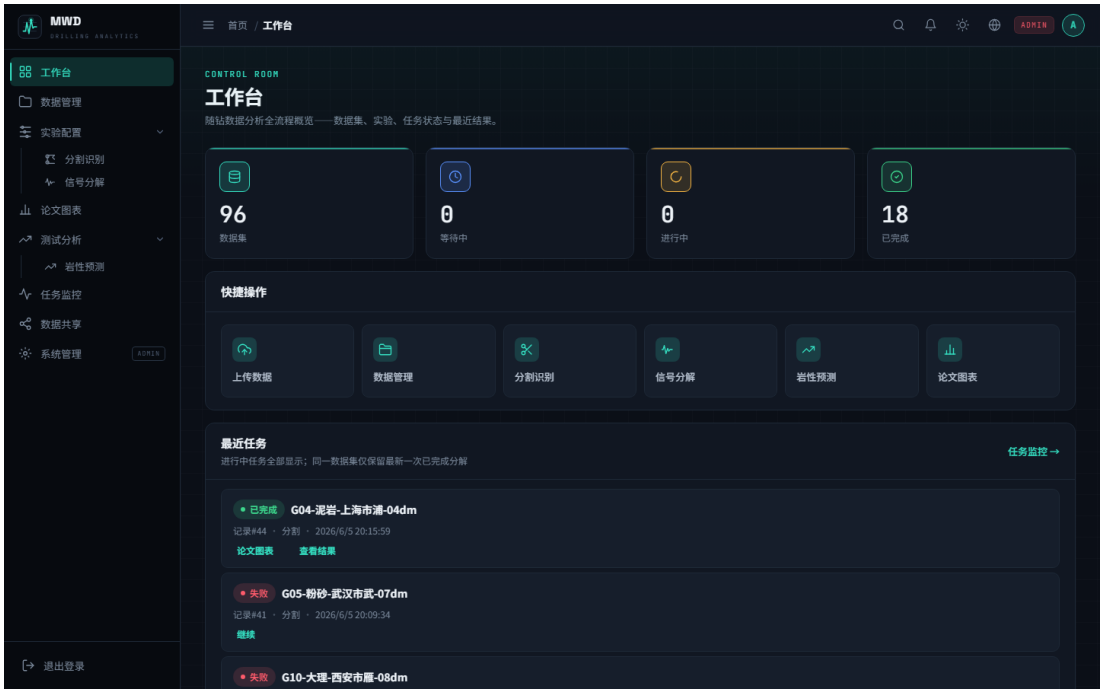


Fig. 2. Dashboard: summary cards, workflow shortcuts and recent tasks.

1.2 Input data format

Uploads are .xlsx files. Each worksheet holds one signal channel of one borehole and is named Drill-{borehole}-{signal}. The first row of a worksheet holds the column names (for example Depth(mm) and Press), the first column is depth and the second the signal value, numerical data begin on the second row, and at least two valid rows are required. Pressure and torque are two independent channels and go in separate worksheets. Metadata entered at upload include the implementing organization, project site, test date, rig model, bit type and diameter, hole depth, control mode, rotational speed, weight on bit, flow rate and rock type. The system detects duplicate files by SHA256 hash to avoid redundant data.

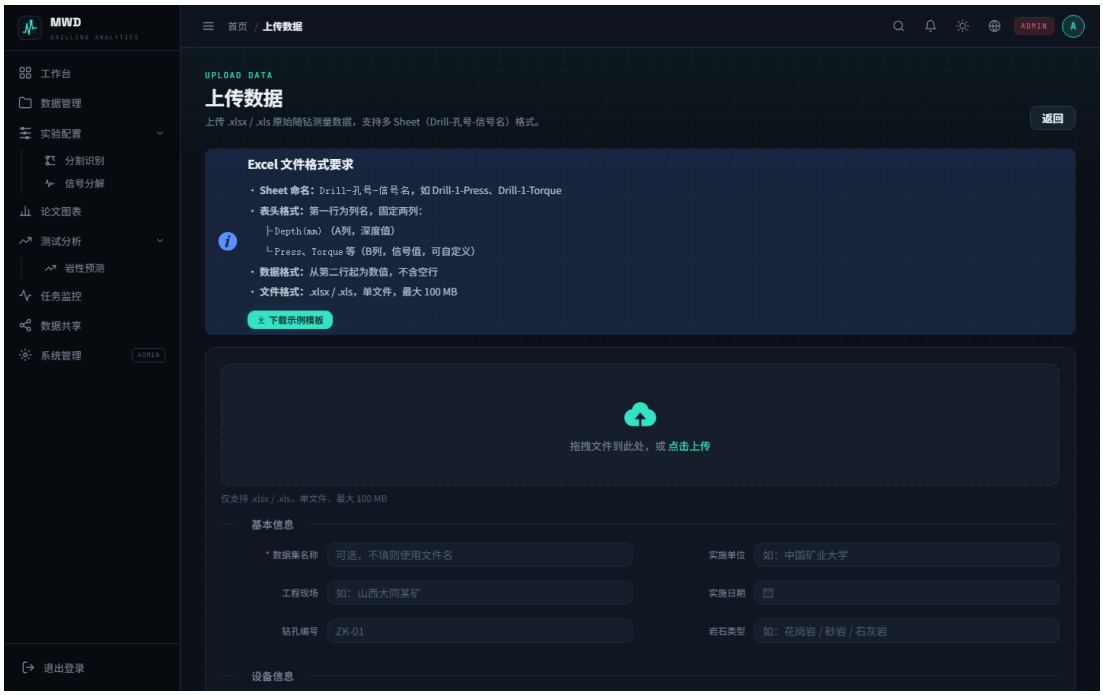


Fig. 3. Data upload: file selection and metadata entry.

1.3 Workflow

The platform divides the method of the main text into three stages — pre-processing, decomposition and prediction — linked in sequence by wizard-style pages, with each submitted task executed asynchronously by a back-end subprocess.

(1) Signal pre-processing. A three-step pipeline turns the raw signal into standardized stable segments ready for decomposition. Data cleaning removes negative values and outliers and applies global Z-score normalization; interpolation offers PCHIP, Akima, cubic-spline and linear methods and resamples the signal to a uniform depth interval (0.05 mm by default); stable-segment partitioning combines global PELT change-point detection (RBF cost, adaptive penalty coefficient) with local sliding-window standard-deviation assessment, automatically dividing the record into four semantic segment types — initial drilling, stable drilling, fluctuation and bit-withdrawal — and locating the step-jump boundaries between segments from Cohen's d effect size. Segment boundaries can be dragged, selected with a brush or split manually, with the result visualized in real time.

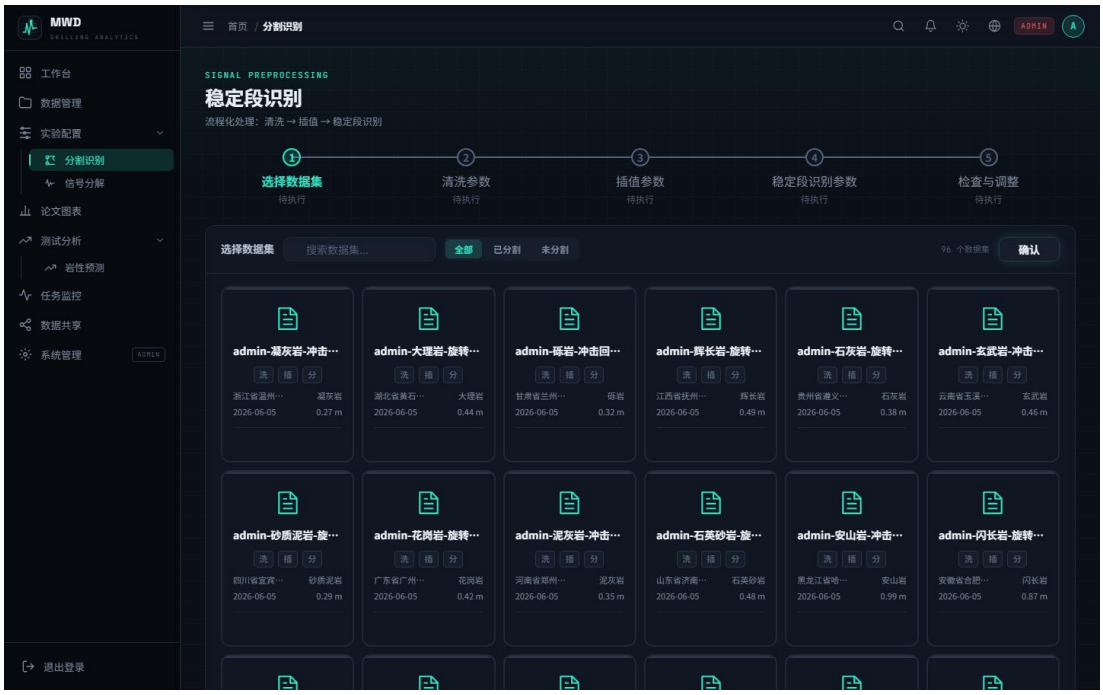


Fig. 4. Three-step pre-processing pipeline and interactive editing of stable segments.

(2) Experiment configuration. The user specifies the control mode (mode A, constant rotational speed; mode B, constant weight on bit) and the range of control parameters, and sets the algorithm parameters. The default algorithm parameters match the configuration used in the main text (Table 1), so submitting them unchanged reproduces the decomposition settings of the paper.



Fig. 5. Experiment configuration.

Table 1. Default decomposition parameters on the platform (identical to the configuration of the main text).

Stage	Parameter	Default
First stage, WPD	Wavelet basis / maximum number of levels / minimum node length	db4 / 5 / 32
	Node discrimination threshold	Otsu, adaptive
Second stage, CWT	Wavelet / number of scales	cmor1.5-1.0 / 128
	Image transform / number of quantile points	FQI / 128
	Soft mask / width coefficient σ	Sigmoid / 1.0
Baseline handling	Baseline subtracted before decomposition and added back to the common component only	Enabled

(3) Signal decomposition. A two-stage funnel-shaped decomposition task is submitted against a saved experiment configuration, and the back end runs four steps in turn. Step 0 reads the stable segments, trims the two signals of the pair to a common length, subtracts the signal baseline and normalizes. Step 1 obtains sub-bands by db4 wavelet packet decomposition, classifies each node as common or differential by Otsu thresholding of its Wasserstein distance, and reconstructs the two coarse components. Step 2 applies the complex Morlet continuous wavelet transform to the coarse components, builds 128-scale FQI images, computes the Wasserstein distance frequency by frequency, performs fine routing in the FFT domain through a Sigmoid soft mask, and merges the result into the final common and differential components. Step 3 computes the cross-borehole validation indices and generates a downloadable validation report.

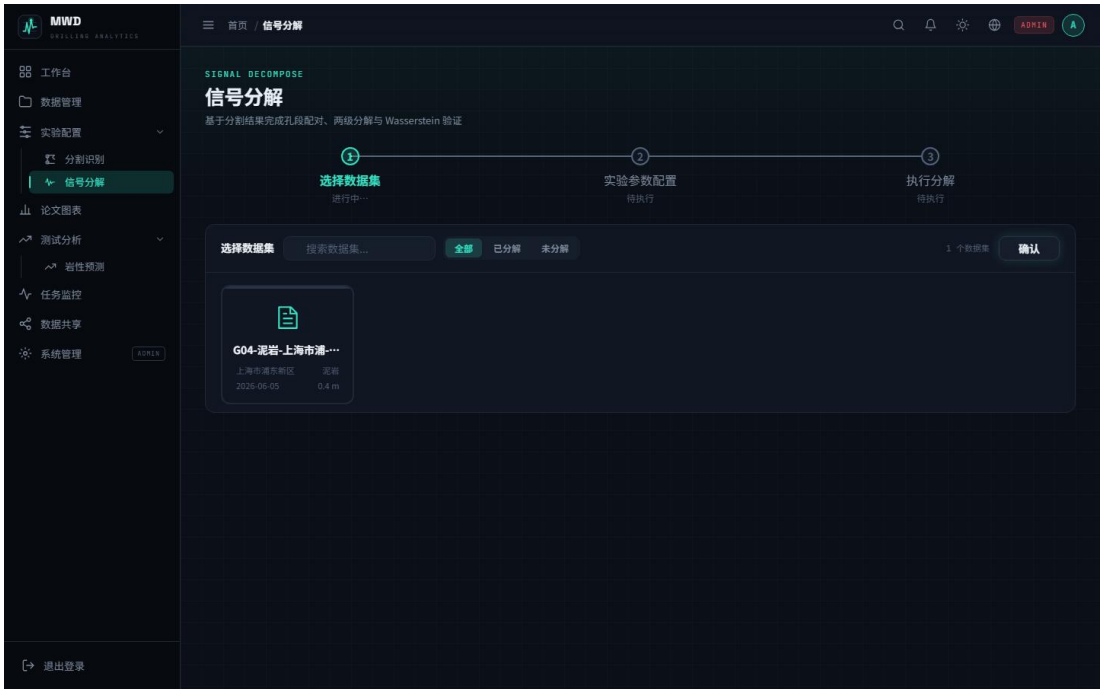


Fig. 6. Submission wizard for a decomposition task.

(4) Task monitoring. The monitoring page lists all asynchronous tasks under four headings — pre-processing, decomposition, figures and prediction — each with a progress bar, a status marker and action buttons. Logs are pushed over WebSocket in real time, so the full output and any error trace can be read in the page, and a failed task can be retried after its parameters are adjusted.

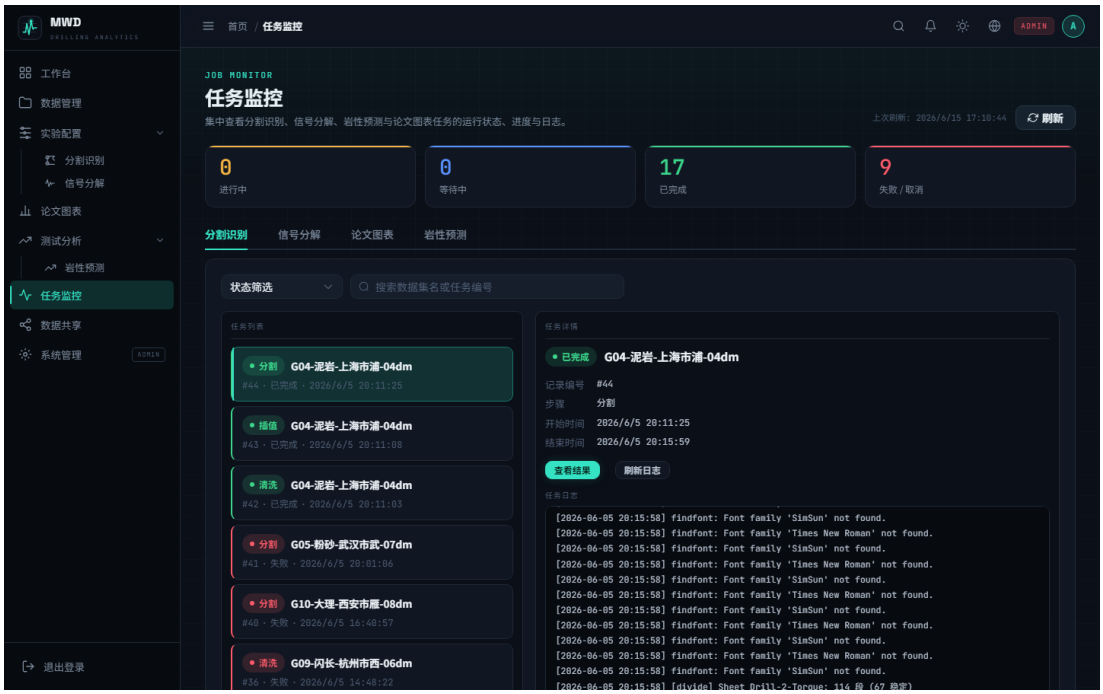


Fig. 7. Task monitoring and real-time logs.

(5) Figure generation. Once a decomposition is complete, figures can be generated in batches for selected borehole pairs and figure types. Seven figure types are available (wavelet packet node decomposition, coarse-stage Wasserstein node discrimination, CWT–FQI time–frequency and frequency-quantile images, the four sub-components, the soft mask, inter-borehole distance maps

and cross-lithology clustering), together with four data exports (tabulated data, curve features, figure source data and the analysis report). Figures are available as PNG, SVG and PDF and can be used directly as manuscript artwork.

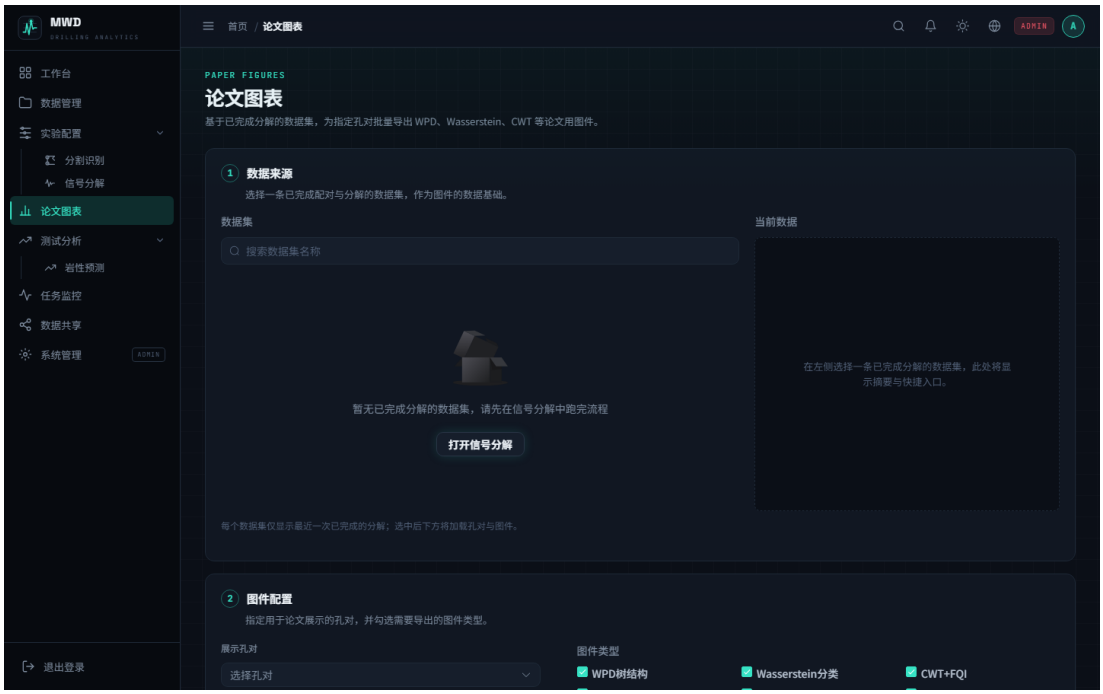


Fig. 8. Batch generation and export of figures.

(6) Lithology prediction. Several datasets can be trained jointly, with either the original signal or the common component as input. Ten features are extracted from each sample (amplitude: mean, root mean square, peak-to-peak; distribution: standard deviation, variance, interquartile range; spectral: dominant frequency, spectral kurtosis, spectral entropy, sample entropy) and, after Z-score normalization, fed to six classifiers: LDA, KNN, SVM-RBF, XGBoost, LightGBM and Random Forest. Training uses a class-stratified 80/20 split with hyperparameters tuned by 5-fold cross-validation and the mean taken over 10 repeats. The platform outputs the accuracy of each model, a feature-importance ranking by ANOVA F value and the confusion matrices, and performs inference on new data in real time.

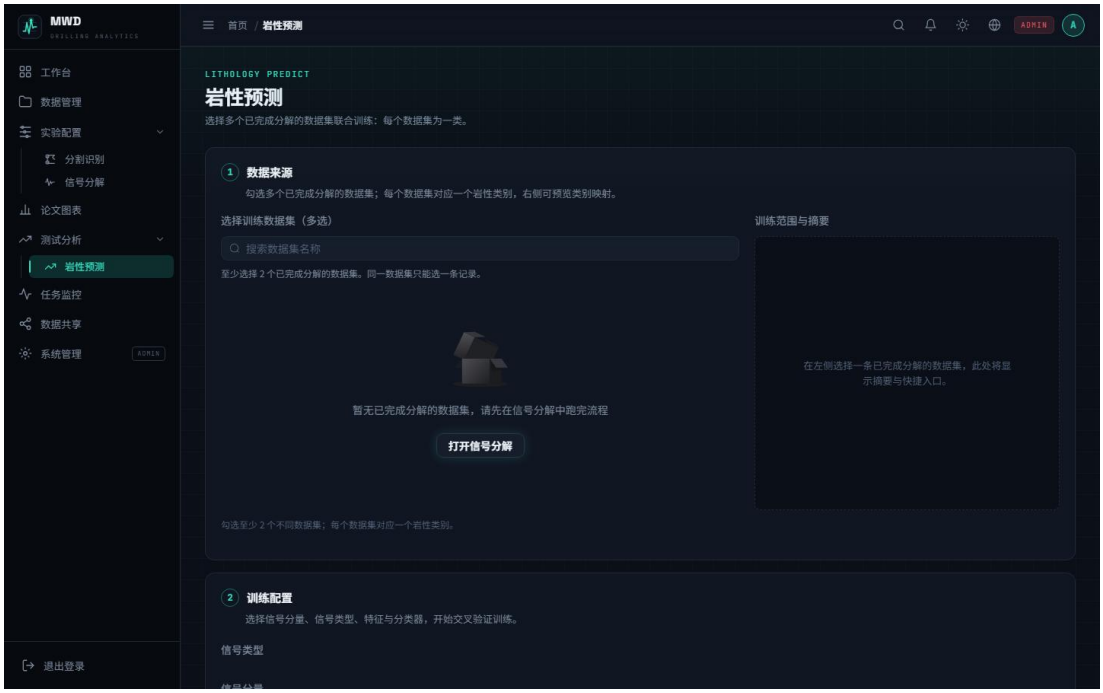


Fig. 9. Lithology prediction: model training and results.

1.4 Worked example: two-stage decomposition of a single-factor rotational-speed pair

The full procedure is illustrated with the single-factor rotational-speed pair of Section 2.1.2 of the main text. Case-1 (ROP 85 mm/min, RPM 100 r/min, DoC 0.85 mm/rev) and Case-2 (ROP 85 mm/min, RPM 250 r/min, DoC 0.34 mm/rev) differ only in rotational speed, all other control parameters and the lithology being identical, so the difference between the boreholes can only be driven by the borehole state.

The steps are: upload the pressure signals of the two boreholes; on the pre-processing page, clean, interpolate and partition the records and take the stable segments; on the configuration page, pair the two boreholes, select control mode A and keep the default algorithm parameters; on the decomposition page, submit the task; once it completes, read the indices on the results page or export the corresponding figures from the figure page. The decomposition result for this pair can be checked directly against the corresponding row of Table 3 of the main text (Table 2), verifying that the platform output agrees with the results reported in the paper.

Table 2. Reference values for the worked example (Case-1 – Case-2, pressure signal), taken from Table 3 of the main text.

Index	Value
Inter-borehole Wasserstein distance of the original signal	2.986 kN
DiC distance / retention	2.785 kN / 93.3%
CC distance / reduction	0.201 kN / 93.3%

1.5 Interface language and common problems

The interface switches between Chinese and English on the fly, menus, forms and figure labels updating immediately without a page reload; dark and light themes are both available, and the setting is kept in the browser's local storage.

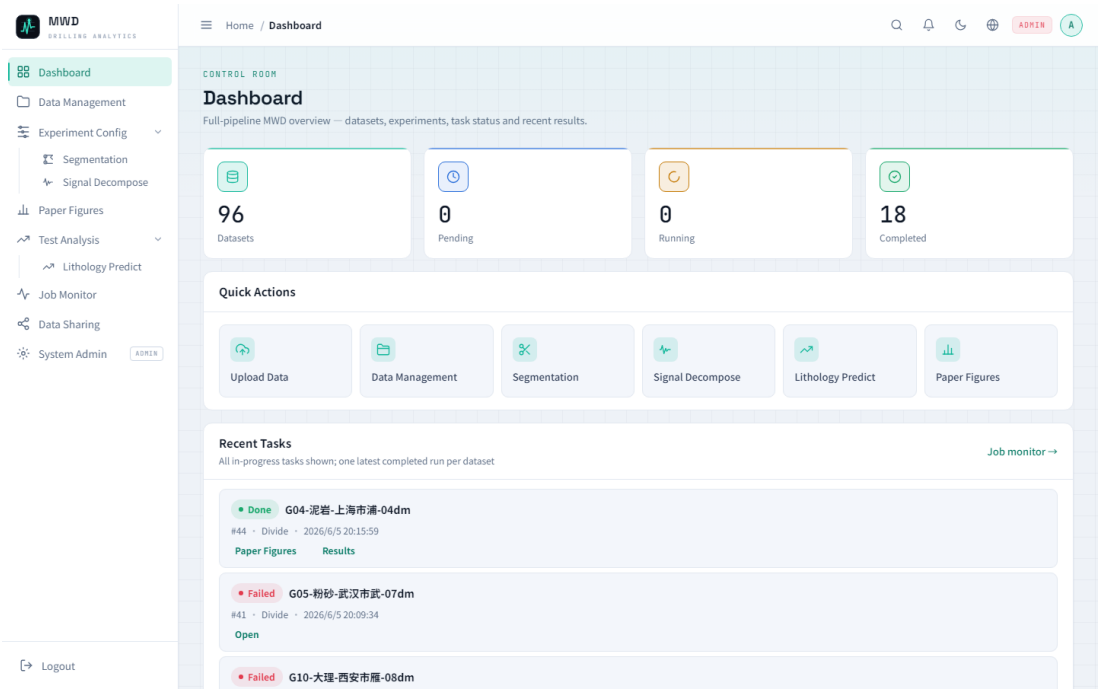


Fig. 10. English interface (dashboard).

If a worksheet name does not follow the Drill-{borehole}-{signal} format, or a required data column is missing, the upload returns a specific error message and the file can be corrected and uploaded again. A failed task can be inspected in full in the log on the monitoring page and retried once its parameters are adjusted. The sign-in token is valid for 24 hours, after which signing in again resumes work. The platform is optimized for desktop use, and interactions such as dragging stable-segment boundaries and zooming figures are best performed on a desktop.